Warmup/Attendance: [Week 5: Friday 9/25/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

**1.** The following problem is taken from Frederick Mosteller's book *Fifty Challenging Problems in Probability*, which is apparently a favorite source for interview questions in quantitative trading and research firms such as Jane Street.

To encourage Elmer's promising tennis career, his father offeres him a prize if he wins (at least) two tennis sets in a row in a three-set series to be played with his father and the club champion alternately: father-champion-father or champion-father-champion, according to Elmer's choice. The champion is a better player than Elmer's father. Which series should Elmer choose: father-champion-father or champion-father-champion? You may treat the result of each game within a set as independent of the other two games.